

PLAYING TOUGHER FOR ONE’S SIZE: CALIBRATED SIZE ABOVE EXPECTED IN THE NHL

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Introduction

Size and hit counts only partly describe how a hockey player competes for the puck and space. A smaller skater may repeatedly battle in traffic or absorb contact, while a larger one may seldom do so. We call this contrast playing tougher for one’s size and develop calibrated size above expected (CSAx) to measure it. We ask whether CSAx agrees with earlier scouting descriptions and relates to NHL deployment, postseason engagement, and next-season continuation.

Methods

We combine NHL events with All Three Zones’ tracked five-on-five puck play from 2021–22 through 2024–25, covering 1,485 forward- and 829 defenseman-seasons. Separate ridge models predict listed height-and-weight size from shared direct measures, such as hits delivered and received, and position-specific indirect measures, such as forward dump-in recoveries and defenseman entry-denial share. We calibrate held-out predictions against listed size and standardize the residual within position and season:

$$\text{CSAx} = \frac{\hat{S} - \hat{m}(S) - \bar{r}}{s_r}, \quad r = \hat{S} - \hat{m}(S).$$

Here S is standardized listed size, $\hat{S}$ is behavior-implied size, $\hat{m}(S)$ is its training-based expectation for that frame, and $\bar{r}$ and s_r summarize held-out residuals by position-season. Higher CSAx indicates a measured physical profile more characteristic of a larger player than expected for that frame. One rater, blinded to identity and CSAx, codes mentions of checking, puck battles, and other active engagement in official NHL draft-year scouting reports published before the studied seasons. Linear models assess special-teams shares and returner ice time; logistic models assess next-season continuation (at least 300 NHL minutes); Poisson models compare hit rates within player-seasons with ice-time offsets. These analyses account for listed size, age and age squared, current-season games dressed and ice time per game, five-on-five points per 60, relative shot attempts, and season. Postseason comparisons use regular-season games excluded from CSAx construction and each team’s first four playoff games, providing a common window regardless of advancement.

Results

Players described as actively engaged in earlier scouting reports average 0.87 CSAx standard deviations higher among forwards and 0.65 among defensemen than peers without such a description (Figure 1).

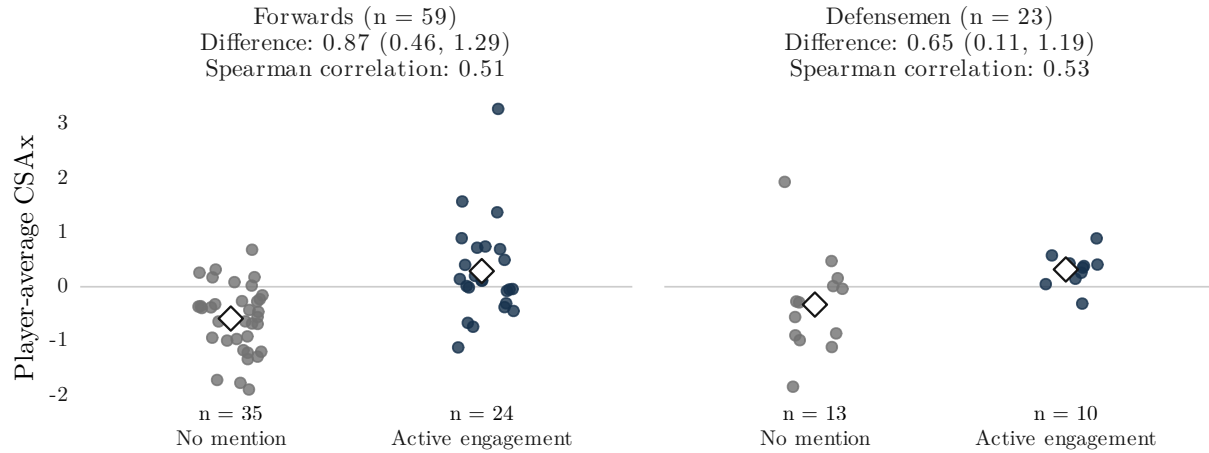


Figure 1. Player-average CSAx by scouting description. Points represent players; diamonds mark group means. Differences show 95% confidence intervals. The high-CSAx defenseman without active engagement description may serve as an example of how our metric can evaluate “tough” players who are sometimes overlooked by scouts due to their relative size; we examine this case further in the full paper.

Teams assign higher-CSAx players in both positions a smaller power-play share and a larger penalty-kill share (Table 1). In the postseason, higher-CSAx forwards show smaller proportional increases in delivered hits; the defenseman contrast remains uncertain. Finally, higher CSAx accompanies a greater chance of next-season continuation, but no clear increase in minutes per game among returners. Thus, continuation need not imply a larger role.

Table 1. Adjusted CSAx +1 versus -1 comparisons (two standard deviations apart; 95% player-clustered confidence intervals in parentheses). Deployment and continuation differences are percentage points; ice time is minutes per game among next-season returners. The postseason value compares postseason-to-baseline hit-rate ratios; below one indicates a smaller proportional increase.

Outcome	Forwards	Defensemen
Power-play share (percentage points)	-1.22 (-1.78, -0.67)	-0.88 (-1.51, -0.25)
Penalty-kill share (percentage points)	1.40 (0.74, 2.06)	1.31 (0.74, 1.89)
Postseason hit-rate ratio	0.78 (0.69, 0.88)	0.90 (0.77, 1.05)
Next-season continuation (percentage points)	2.92 (0.03, 5.81)	4.23 (0.60, 7.86)
Ice time (min/game)	-0.18 (-0.36, 0.00)	-0.14 (-0.43, 0.14)

Conclusion

CSAx adds a frame-relative perspective to player evaluation alongside size, hits, and scoring. Its agreement with earlier scouting descriptions and its links to deployment and next-season continuation suggest value in assessing specialized roles. Selectively tracked games, one scouting rater, and weak defensive size prediction in 2024–25 limit generalization.